

SECURITY ASSESSMENT REPORT

Automated Incident Response & Vulnerability Scan Analysis

Target IP Address: 192.168.29.79

Scan Timestamp: 2026-08-12T10:58:30Z

Report Generated: 2026-08-12 10:58:43 UTC

OVERALL THREAT RISK LEVEL

MEDIUM

1. Executive Summary & AI Analysis

Risk Overview

The overall risk rating for this host is MEDIUM. The presence of open ports and services, particularly those related to Windows RPC and SMB, poses a risk of exploitation, especially considering the outdated CVE found for the microsoft-ds service. However, the lack of high-severity CVEs and the fact that the most critical services are not directly exposed to the internet (assuming a typical network setup) mitigate the risk.

Prioritised Vulnerabilities

- **CVE-2002-0597 (MEDIUM):** This vulnerability is dangerous because it allows remote attackers to cause a denial of service (DoS) by sending malformed data to the microsoft-ds port 445, potentially leading to CPU or memory exhaustion. It is moderately easy to exploit since it requires sending specific malformed data but does not need authentication. The exploitability is somewhat mitigated by its age and the assumption that more recent Windows versions may have mitigations in place.
- **Open Services:** While not vulnerabilities per se, the open ports for msrpc, netbios-ssn, and microsoft-ds services pose a risk due to their potential for exploitation in lateral movement or as part of a larger attack strategy.

Critical Attack Vectors

The most likely paths an attacker would take to compromise this host involve exploiting the open services for discovery and lateral movement:

- **T1046 (Network Service Discovery):** An attacker could use the open ports (135, 139, 445, 5000) to discover available network services, which could be the first step in identifying potential vulnerabilities.
- **T1021.002 (Remote Services: SMB/Windows Admin Shares):** Specifically, the microsoft-ds service on port 445 could be targeted for lateral movement within the network, potentially allowing an attacker to move laterally if they gain initial access.

Immediate Remediation Steps

- **Disable or Restrict Access to SMB:** Immediately restrict access to the microsoft-ds service (port 445) to only necessary IP addresses or disable it if not required. This can be done through Windows Firewall or by disabling the SMB service if it's not necessary.
- **Update and Patch:** Ensure the system is updated with the latest security patches, even though no specific high-severity patches were mentioned, keeping the system up to date is crucial.

- **Limit Exposure of HTTP Service:** If the HTTP service on port 5000 is not necessary, consider disabling it or restricting access to it. If it's a development server, ensure it's not exposed to unnecessary networks.
- **Monitor for Anomalies:** Implement monitoring for anomalies on these open ports, especially for signs of exploitation attempts against the known CVE or unusual service discovery activities.

Long-Term Hardening Recommendations

- **Segment Network:** Implement network segmentation to limit lateral movement in case an attacker gains access to the network.
- **Regular Vulnerability Scans:** Conduct regular vulnerability scans to identify and address any new vulnerabilities that may arise.
- **Implement Firewall Rules:** Tighten firewall rules to only allow necessary incoming and outgoing traffic.
- **Secure Remote Access:** Ensure all remote access to the system, especially for administrative tasks, is secured with strong passwords, multi-factor authentication, and encrypted connections.
- **Consider Replacement of Outdated Services:** If possible, consider replacing or upgrading services that are outdated or known to have security issues with more secure alternatives.

2. Open Ports & Detected Services

Port	Service	Version	Risk
135/tcp	msrpc	Microsoft Windows RPC	LOW
139/tcp	netbios-ssn	Microsoft Windows netbios-ssn	LOW
445/tcp	microsoft-ds	N/A	HIGH
5000/tcp	http	Werkzeug httpd 3.1.8 Python 3.14.3	LOW

3. Discovered Vulnerabilities (CVEs)

CVE-2002-0597 — CVSS Score: 5.0 | Published: 2002-06-18

MEDIUM SEVERITY

LANMAN service on Microsoft Windows 2000 allows remote attackers to cause a denial of service (CPU/memory exhaustion) via a stream of malformed data to microsoft-ds port 445.

4. MITRE ATT&CK; Mappings

Port	Technique ID	Technique Name	Tactic
135	T1046	Network Service Discovery	Discovery
139	T1046	Network Service Discovery	Discovery
445	T1021.002	Remote Services: SMB/Windows Admin Shares	Lateral Movement
5000	T1046	Network Service Discovery	Discovery

5. Immediate Remediation & Hardening

- **Patch & Update Software:** Immediately apply vendor security updates for all identified outdated services and libraries.
- **Enforce Strict Firewall Policies:** Restrict inbound network access to essential ports only; block unused open ports.
- **Implement Strong Authentication:** Enforce multi-factor authentication (MFA) and disable password-based SSH authentication in favor of key pairs.
- **Continuous Monitoring & Auditing:** Enable continuous log monitoring, intrusion detection systems (IDS), and regular automated vulnerability assessment.